

Automata, Logic and Games HW 1

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Assignment 1

Let

$$Inf(Z) = \forall x \exists y (x \leq y \wedge Z(y)); \text{ Par}(X, Y) = (Inf(X) \leftrightarrow Inf(Y)),$$

$$\varphi'(X, Y) = \varphi(X, Y) \wedge \text{Par}(X, Y).$$

Claim: there exists a uniformisation $\psi(X, Y)$ of $\varphi(X, Y)$ with $\psi \Rightarrow \text{Par}$ iff the sentence

$$\forall X (\exists Y \varphi(X, Y)) \Rightarrow \exists Y \varphi'(X, Y)$$

holds.

($\Rightarrow$) If such ψ exists, the statement is immediate as ψ is a uniformisation of φ .

($\Leftarrow$) For fixed word and X , consider all Y with $\varphi'(X, Y)$. View Y as a 0, 1-labelling. Let

$$\psi(X, Y) = \varphi'(X, Y) \wedge \forall Y' (Y' < Y \rightarrow \neg \varphi'(X, Y')),$$

where $Y' < Y$ is the lexicographic order induced by the labeling. Essentially, ψ chooses the lexicographically minimal witness. At the same time, $\psi \subseteq \varphi' \subseteq \varphi$ and φ projects onto φ' (by the assumption above) as well as φ' projects onto ψ , therefore ψ is a uniformisation of φ . $\psi \Rightarrow \text{Par}$ because φ' does.

Therefore the problem is equivalent to truthfulness of the MSO sentence. Since MSO over ω -words is decidable, our problem is decidable.

Assignment 2

We view a word $w \in A^*$ together with sets $Y_1, \dots, Y_n$ as a word

$$\hat{w} \in B^*, \quad B = A \times \{0, 1\}^n,$$

Every pMSO formula $\psi(X_1, \dots, X_n)$ over A can be translated into a sentence $\hat{\psi}$ over B such that

$$(w, Y_1, \dots, Y_n) \models \psi \iff \hat{w} \models \hat{\psi}.$$

Let $u =_k v$ on B^* iff u and v satisfy the same pMSO sentences of rank $\leq k$. It is obviously an equivalence relation.

From lectures, we know that the map $\hat{w} \rightarrow$ all pMSO sentences of rank $\leq k$ that w satisfies is a homomorphism onto a finite monoid. Therefore, $=_k$ has finite index and

$$u =_k v \Rightarrow uc =_k vc$$

- has the property of right congruence.

Let $Q = B^*/=_k$ - finite since the index of the relation is finite, and

$$\mathcal{A} = \langle B, Q, q_I, \delta \rangle, \quad q_I = [\varepsilon]_k, \quad \delta([u]_k, c) = [uc]_k.$$

For an (n, k) -formula ψ let

$$F_\psi = \{ [u]_k \in Q : u \models \hat{\psi} \}.$$

Now for any $w \in A^*$ and $Y_1, \dots, Y_n$

$$\mathcal{A} \text{ with } F_\psi \text{ accepts } (w, Y_1, \dots, Y_n) \iff [\hat{w}]_k \in F_\psi \iff \hat{w} \models \hat{\psi} \iff (w, Y_1, \dots, Y_n) \models \psi.$$

Which is precisely what we had to show.